Luiz Takahashi

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Education

Washington State University **Pullman, WA**
B.S. in Computer Science, GPA: 3.84/4.0 **Expected May 2027**
President's Honor Roll (4 semesters). Coursework: Machine Learning, Data Mining, Algorithms, Linear Algebra.

Research Experience

Independent Researcher, Computational Biology (single-cell / neural organoids) **2025 – Present**
Self-directed; sole-author preprint (Zenodo, 2026).

- Conducted a single-author study of the off-target compartment of human neural organoids: mapped a 1.77M-cell organoid atlas onto a harmonized 2.6M-cell fetal-brain reference (scVI / scANVI / scArches), defining calibrated off-manifold and label-entropy scores that resolve a six-state taxonomy of off-target cells.
- Established an *identifiability boundary*: proved a standard contrastive disentanglement model cannot separate lineage from in-vitro state (an adversary recovers cell source at >0.998 accuracy regardless of latent capacity), then recovered confound-invariant geometry via sliced-Wasserstein cancellation, validated with a confound-controlled cross-atlas test.

Undergraduate Research Assistant, CASAS Smart Home Lab & Neuropsychology and Aging Lab **Aug 2025 – Present**
Washington State University. Advisors: Dr. Diane J. Cook & Dr. Maureen Schmitter-Edgecombe. Funded by NIH/NIA.

- Developed a room-aware Kalman filter for indoor localization from sparse smart-home sensors monitoring 35 older adults (ages 60–90+, 4 days to 22 months); reduced localization error by 18% with 92% statistical consistency (vs. 65% baseline) across 2M+ sensor events.
- Found cognitive accuracy significantly associated with behavioral variety ($r = -0.43$, $p = 0.015$), driven almost entirely by MCI participants ($r = -0.93$, $p = 0.002$); showed the effect dissolves under a noise-multiplied outcome, isolating where the signal lives.
- Built a LASSO classifier (healthy vs. impaired, AUC = 0.78) identifying spatial entropy ($\beta = -0.42$) and night-activity fraction ($\beta = +0.38$) as clinically interpretable biomarkers consistent with sundowning.

Independent Researcher (Columbia University / Wen Lab collaboration), Multi-View Interaction Hierarchy **2025 – Present**

- Developed a hierarchy for multi-view interaction detection with completeness guarantees; proved that the common view-disagreement heuristic cannot detect interactions, and bounded the interaction order recoverable at a given sample size via a Hoeffding functional decomposition.
- Demonstrated on a 30,000-person GNPC proteomic dataset (9 organ biological-age gaps, 18 disease outcomes) that a seemingly significant cross-organ interaction effect is a between-cohort batch artifact, reproducing the Simpson's paradox from first principles.

Research Assistant, Center for Artificial Intelligence (C4AI), University of São Paulo **May – Aug 2025**
HarpIA Project. Advisor: Dr. Fabio Cozman.

- Developed SQL Component Match, a semantic natural-language-to-SQL evaluation metric using AST analysis ($\sim$4,000 lines of Python): 95% semantic accuracy vs. a 70% Spider baseline with $O(n^{1.08})$ scalability; built a 5,686-line test suite (234 methods) with multi-stage normalization, a DNF circuit breaker, and Unicode homograph detection.

Publications & Presentations

- **L. Takahashi.** *NullState: a reference-anchored framework for detecting, diagnosing, and quantifying the off-target compartment of human neural organoids.* Preprint, Zenodo, 2026. doi:10.5281/zenodo.21459720. (Sole author.)
- **L. Takahashi**, J. Wen. Multi-view interaction hierarchy for health screening. *In preparation* (target: AISTATS 2027).
- **L. Takahashi**, D. J. Cook, M. Schmitter-Edgecombe. Day-similarity structure in older adults. *In preparation* (target: journal).
- **L. Takahashi**, D. J. Cook, M. Schmitter-Edgecombe. Room-Aware Behavioral Profiling for Cognitive Health Monitoring in Smart Homes. *Gerontechnology* 25(Suppl.), 2026 (ISG World Conference, Vancouver). doi:10.4017/gt.2026.25.2.1421.3.

Teaching

Teaching Assistant, Washington State University **2025 – Present**

- **Program Design & Development (C/C++):** lead a weekly 3-hour lab for 15+ students; mentor on data structures, algorithms, and debugging during office hours.
- **Data Mining:** sole teaching assistant for the course; graded assignments and exams for 50+ students.

Technical Skills

Languages: Python (advanced), C (proficient), C++, SQL, C#, Bash **ML/AI:** PyTorch, scikit-learn, scvi-tools/Scanpy, Kalman filtering, LASSO/Ridge, KNN, neural networks
Data: Pandas, NumPy, SciPy, Matplotlib, Seaborn, sqlglot, time-series analysis, synthetic data generation **Tools:** Git, Docker, Linux, LaTeX, Jupyter